

youkined from <https://github.com/Nicicalu/OST-CN2-Spik>

OSPF (IGP)
Equal Cost MultiPath (ECMP) load balancing, fast convergence, widely used. Uses IP Port 89 on Layer 3. Uses 224.0.0.5 (all routers), 224.0.0.6 (all DR,BDR)
DR Election: Highest interface priority → if tie, highest Router ID. Priority 0 = ineligible for DR/BDR. No preemption. BDR = same rules, second best candidate
ROUTER OPERATION
– Establish neighbor adjacencies and exchange LSAs
– Build the Link-State Database (LSDB)
– Run Dijkstra's SPF algorithm:
– Intra-area change: SPF recalculation
– Inter-area change: No SPF needed → ABR handles updates
– Build the routing table from SPF results
AREAS
– AS split into areas (sub-domains), each with a 32-bit Area ID (e.g., 0.0.0.0 = Area 0)

Backbone Area (Area 0): Core of the OSPF domain (must exist). Must connect to all other areas (directly or via virtual links). Must be contiguous (no disjointed segments). Should not contain end-user networks

Non-Backbone Areas: Connect end users and local resources. All inter-area traffic must transit the backbone

ROUTER TYPES
Area Border Router (ABR): Connects two or more OSPF areas; must have interface in backbone. 1 OSPF DB per Area
Internal Router: All interfaces belong to the same area (non-backbone)
Backbone Router: At least one interface in Area 0. Includes ABRs and routers internal to the backbone
AS Boundary Router (ASBR): Connects to external AS/Network (e.g., BGP). Advertises external routes into OSPF. Can be in backbone or non-backbone area

DESIGN RULES
Rule 1: Backbone (Area 0) must be contiguous → no partitions allowed
Rule 2: Every non-backbone area must connect to Area 0

LSA TYPE OVERVIEW
Type 1 – Router-LSA: Sent by all routers, lists directly connected links (so all outgoing interfaces with state and cost) (intra-area only)
Type 2 – Network-LSA: Sent by DR, lists all routers and DR in broadcast/multi-access network (intra-area only)
Type 3 – Summary-LSA: Sent by ABRs, advertises networks from other areas, flooded in all the areas that are not "totally stubby" (inter-area)
Type 4 – ASBR-Summary: Sent by ABRs, advertises path to ASBRs (inter-area)
Type 5 – AS-External-LSA: Advertises external routes (e.g., BGP); flooded to all non-stub areas – only normal areas
Type 7 – NSSA External: Like Type 5 but used inside NSSA; converted to Type 5 by ABR

AREA TYPES (ALLOWED LSA TYPES)	
Standard (1,2,3,4,5): Normal	Not-So-Stubby Area – NSSA (1,2,3,7):
Stub Area (1,2,3):	(1,2,3,7):
– Blocks external LSAs (Type 5)	– Like stub area, but allows one ASBR inside
– ABR injects a default route (0.0.0.0)	– External routes use LSA Type 7 (converted to Type 5 by ABR)
– Supports LSAs Types 1–3	
Totally Stubby Area (1,2):	Totally NSSA (1,2,7):
– Blocks external (Type 5) and summary (Type 3/4)	– Like NSSA but blocks Type 3 so gets default route from ABR
– Only allows default route from ABR	

PACKET TYPES
Type 1 – Hello: Used to discover, maintain, and verify neighbors; forms adjacencies. Also for election of DR,BDR in broadcast networks. Contains network mask(of sending routers' interface), Hello interval(p2p,broadcast: default=10s), Options, Priority(for election), Router dead interval(default=40s), DR/BDR/ID, Neighbors
Type 2 – Database Description (DBD/DD): Exchange summaries of LSAs during adjacency formation (headers only). Contains Interface Max, MTU, Options, I/M/MS bits (Initial, More, Master-slave bit), DD Sequence num, LSA Header
Type 3 – Link State Request (LSR): Sent when a router needs specific LSAs listed in the DBD. Contains Link State Type (router/network), Link State ID, Advertising Router (sender address)
Type 4 – Link State Update (LSU): Used to flood new or updated LSAs. Contains Number of LSAs, full LSAs information
Type 5 – Link State Acknowledgment (LSAck): Confirms receipt of LSAs to ensure reliable flooding. For this you send LSAck or implicitly by sending LSU with same info back. Many acks may be grouped together to a single LSAck.

SUB-PROTOCOLS
Hello Protocol:
– Used for neighbor discovery and parameter negotiation.
– Maintains logical adjacencies on P2P, P2MP, and virtual links.
– Elects DR/BDR on broadcast and NBMA networks.
– Continuously sends hello packets to maintain bidirectional connectivity; failure to receive = neighbor down (in agreed router dead interval at initialization)
Database Sync Protocol:
– Syncs LSDB using Database Description (DBD) packets with only LSA headers.
– Uses I-bit (initial), M-bit (more), and MS-bit (master/slave).
– ExStart: Bi-direction; highest Router-ID = master. Determine initial seq nr
– Exchange: Exchange of DBD packets (LSA headers).
– Loading: Missing LSAs are requested.
– Full: Databases fully synchronized.

OSPF ROUTING AND ECMP
– Each router runs Dijkstra per area; link cost = metric from LSAs (1–65535)
– OSPF prefers more specific match (CIDR) and if then still multiple: intra-area > inter-area > external
– Routes added to RIB/FIB based on computed next hops
– ECMP: Modified Dijkstra supports Equal-Cost MultiPath if multiple paths have same cost → routes added with multiple next-hops for load balancing

OSPF ROUTE SELECTION
Intra-Area (O): Source and dest in same area; routes from Type 1 and 2 LSAs
Inter-Area (O IA): Source and dest in different areas within same AS; via Type 3 LSAs through backbone

External (E1/E2): Dest outside AS; info injected by ASBR via redistribution
– E1: Total = external + internal OSPF cost
– E2: Only external cost (default)
Preference order: More specific route > Intra-area > Inter-area > E1 > E2
Cost calculation: Cost = Reference Bandwidth/default 100 Mbps/Interface Bandwidth
IS-IS (IGP)
CLNS – CONNECTIONLESS NETWORK SERVICES
CLNS: ISO Layer 3 datagram service; supports CLNP ES-IS, IS-IS.
CLNP: Connectionless Network Protocol, similar to IP; used in ISO stack (EtherType 0xFEEF).
IS-IS: Link-state routing protocol (Layer 3); forms adjacencies with ES-IS; designed for CLNP but extended (Integrated IS-IS) to support IP.
Integrated IS-IS: Allows IP routing with IS-IS; used widely by service providers even without CLNP.

NET ADDRESSING (TYPE OF NSAP ADDRESS)
NSAP: Network Service Access Point
NET = Network Entity Title: Unique router identifier in IS-IS
Format: 49.AAAA.BBBB.BBBB.BBBB.00
49.AAAA...: Area ID (variable length)
BBBB.BBBB.BBBB: System ID (usually 6 bytes = unique router ID)
00: N-selector (NSEL) (always 00 for routers)
System ID: Must be unique per router (often based on lo-IP/MAC)
Example: 49.0001.1921.6800.1024.00 based on IP 192.168.1.24
Area ID: Used for routing hierarchy (like OSPF areas)
IS-IS PACKET TYPES (ALL IN L1 OR L2)
IIH (Hello): Builds and maintains adjacencies; includes system ID, holding time, prio
– Built from 3 functions: discover, build, maintain
– Interval: 10s (default); DIS sends every 3.3s on LANs
– Multipier: Missed Hello limit → Holdtime + interval * Multiplier(default=3)
Multicast:
– 01-80-c2-00-00-14 (All L1ISs)
– 01-80-c2-00-00-15 (All L2ISs)
LSP (Link State PDU): Contains topology info including prefixes with costs; flooded throughout the area → similar to OSPF LSA Type 1
CSNP (Complete SNPP): Sent by DIS; lists all known LSPPs (used for database sync)

PSNP (Partial SNPP): Used to request missing LSPs or acknowledge received LSPs
IS-IS Packet Structure: Common header + TLVs
ROUTER TYPES
L1 Router: Only within one area; no inter-area routing
L2 Router: Backbone router; routes between areas
L1/L2 Router: Acts as both; separates databases, redistributes between levels
BROADCAST/MULTI-ACCESS LINKS: DIS – DESIGNATED IS
– Required on broadcast links (no DIS on p2p)
Sends periodic CSNPs to ensure DB sync, creates pseudonode LSP
No backup DIS in IS-IS

DIS ELECTION
– 1) Highest interface priority (0–127) Cisco default = 64
– 2) Highest SNPA (MAC-Address)
Preamption: Enabled – higher prio router automatically takes ver the DIS Role
POINT-TO-POINT LINKS (NO DIS)
CSNP: Sent once at adjacency startup
LSP: Advertises topology changes (link-state info)
PSNP: Acknowledges received LSPs or requests missing ones
PATH SELECTION
Path Selection Order (in IS-IS) – Lower Metric better:
– L1 intra-area routes
– L2 intra-area routes
– Leaked L2 → L1 (internal metric)
– L1 external (external metric)
– L2 external (external metric)
– Leaked L2 → L1 (external metric)
LEVEL 1 ROUTING
– Intra-area routing only (like OSPF intra-area)
– L1 routers use the closest L1/L2 router for inter-area traffic
L1/L2 routers:
– Do not advertise L2 routes into the L1 area (unless route leaking is active)
– Set Attached bit to signal L2 connectivity to backbone
– L1 routers install a default route to nearest L1/L2
– L1 area like OSPF Totally Stubby Area
– Distribution Bit: Set to 'up' (1) on L2 → L1 leaks; blocks re-advertisement L1 → L2.
– Route-Leaking injects a more specific route into L1 to improve routing

LEVEL 2 ROUTING		
– Routing between areas (inter-area)		
– L1/L2 routers inject L1 routes into L2 topology		
– L1 routes are redistributed into L2 with L1 metric preserved in L2 LSP		
IS-IS VS OSPF		
Feature	IS-IS	OSPF
Layer	L2 (CLNS)	L3 (IP, proto 89)
Encapsulation	No IP, uses TLVs	IP packets
Hello Type	IIH	Hello packet
Area Model	L1/L2	Backbone + Areas
Metric	Cost (default 10)	Cost (bandwidth)
Router ID	System ID (6B)	32-bit Router ID
Adj. Types	L1, L2, L1/L2	DR/BDR, P2P
LSDB	Per level (L1/L2)	Per area
Scaling	Large-scale ISP core	Enterprise/campus
Routing info	TLVs (flexible)	Fixed LSA types
BGP (EGP)		
CONFIG		
– next-hop-self fixing iBGP: neighbor (neighbor IP) next-hop-self (when overriding)		
BGP SESSIONS		
– Point-to-point adjacencies between BGP routers		
– iBGP: Between routers in the same AS, AD=200, more trusted (lower security overhead)		
– eBGP: Between routers in different ASes, AD=20, stricter policy enforcement		
AUTONOMOUS SYSTEM NUMBERS (ASN)		
– Unique ID for each AS; required for Internet routing with BGP		

– Private ranges:
– 64512–65535 (legacy 16-bit)
– 4200000000–4294967294 (32-bit)
BGP PEERING / NEIGHBORS
– Two routers with a BGP TCP session (port 179) are called peers or neighbors
– Each BGP router is a BGP speaker
– BGP exchanges routing info between ASes (loop-free, policy-based)
– Supports CIDR, route aggregation; decisions based on policies/rules
PATH ATTRIBUTES
Used for route control and policy enforcement
Well-known mandatory: Always present (e.g., AS-Path, Origin, Next Hop)
Well-known discretionary: Optional but recognized by all (e.g., Local Pref, Atomic Aggregate)
Optional transitive: Passed between ASes (e.g., Community, Aggregator)
Optional non-transitive: Not passed across ASes (e.g., MED, Weight, Originator ID, Cluster ID/list)
NLRI: Routing table info: prefix, prefix length, and associated path attributes
LOOP PREVENTION
BGP uses AS-Path (list of ASNs) to detect loops. If a router sees its own ASN in a received route, it discards it.
AS-Override: Allows reuse of same ASN across different customer sites (e.g., Swisscom); rewrites ASN to avoid loop detection

BGP MESSAGES
OPEN: Establishes session; includes version, ASN, Hold Time, BGP Identifier, optional params
– Hold Time: Heartbeat in seconds (default 180s, Cisco), reset by KEEPALIVE/UPDATE; 0 = session down
– BGP Identifier: 32-bit Router-ID, manually set or highest loopback/active IP; used for loop prevention
KEEPALIVE: Sent every 1/3 of Hold Time (default 180s); ensures neighbor liveness (BGP doesn't rely on TCP keepalive)
UPDATE: Advertises new routes, withdraws old ones, or both; includes NLRI (prefix + path attributes); can act as KEEPALIVE
NOTIFICATION: Sent on session error (e.g. hold timer expired); terminates session immediately
BGP NETWORK STATEMENTS
– Purpose: Advertise specific prefixes to BGP peers (does not activate interfaces)
– Prefix must exist exactly in the RIB (from static, connected, or learned route)
– Attributes (e.g., origin, next-hop, MED) depend on how the route exists in RIB
– BGP advertises only the best path for a prefix to peers, even if multiple exist
BEST PATH CALCULATION
– BGP maintains all received paths per prefix but advertises only the best one
– Best path is installed in RIB; recalculated on:
– Next-hop reachability change
– Interface failure to eBGP peer
– Redistribution change
– New/withdrawn path received
– Influence:
– Outbound BGP policy → inbound traffic behavior
– Inbound BGP policy → outbound traffic behavior

BGP BEST PATH SELECTION (IN ORDER):
1) Prefer highest Weight (Cisco-specific, local to router)
2) Prefer highest Local Preference (global within AS)
3) Prefer routes originated by the router (only small i in path, NH 0.0.0.0)
4) Prefer shorter AS path (only length is compared)
5) Prefer lowest origin type: IGP < EGP < Incomplete (I on origin)
6) Prefer lowest MED (Multi-Ext Discriminator) (also called metric)
7) Prefer external (eBGP) over internal (iBGP)
8) For iBGP: prefer path with lowest IGP metric to next-hop
9) For eBGP: Prefer oldest (more stable) path
10) Prefer lowest BGP router ID
11) Prefer path from lowest neighbor IP address
ROUTE FILTERING
– Filters control which routes are received/advertised
– Used for security, traffic shaping, memory optimization
– Tools: prefix-list (IP), filter-list (AS-path), route-map (flexible match/set)
BGP COMMUNITIES
– 32-bit optional, transitive tag (e.g. ASN-value, 65000:100)
– Used to mark routes for policy control across ASes
– Can be added, modified, or removed at each hop
iBGP SCALABILITY
– iBGP does not re-advertise routes between iBGP peers → full mesh required (causing no loop prevention)
– Session count = $\frac{n(n-1)}{2}$ (e.g., 5 routers = 10 sessions, 10 = 45)
ROUTE REFLECTORS (RR)
– Solves iBGP full-mesh scaling by allowing selective route reflection
– Clients only peer with RR; unaware they're clients
– RR Rules:
– From non-client → advertise to clients only
– From client → advertise to all (clients & non-clients)
– From eBGP peer → advertise to all (clients & non-clients)
– Only the RR needs special config – clients remain unaware of route reflection. This eliminates the need for full iBGP mesh.

PEERING VS. TRANSIT
Transit: ISP provides full reachability (paid relationship)
Peering: ISPs exchange selected routes; equal relationship, usually unpaid
AS Path Filtering: to avoid getting transit; ip as-path access-list 10 permit ^ \$
INTERNET EXCHANGE POINT (IXP)
– Facility where networks exchange traffic via BGP peering
– Reduces transit costs, latency, and offloads upstream links
PUBLIC PEERING
– Members peer via shared switch fabric and a route server
– Route server distributes routes but stays out of data path (NEXT_HOP unchanged)
– Minimal policy control; one BGP session to route server
– Simplified setup (one legal contract)
PRIVATE PEERING
– Direct BGP sessions between two parties (1:1)
– May use public or private interconnects
– Full policy control per neighbor
– Requires one session and legal contract per peer
Examples: Equinix, SwissIX (non-profit)
ENTERPRISE CONNECTIVITY OPTIONS
Single-Homed: One ISP, one link (BGP or static); simple but no redundancy

Dual-Homed: One ISP, two links (or routers); redundancy within same provider
Multihomed: Multiple ISPs; improved redundancy and routing control, but avoid being a transit – advertise only customer-owned prefixes
Dual-Multihomed: Multihomed, but two links per ISP
TRAFFIC ENGINEERING (TE)
Outbound TE (Local Pref): Set higher local pref to prefer exit path; affects outbound traffic; (highest wins)
Inbound TE (MED): Signal entry preference with MED; lowest wins; only works if peer honors it
Inbound TE (AS-Path Prepending): Add own ASN multiple times on backup path; shortest AS-path wins
TE Limitation: Affects only routes outbound (e.g. local pref); inbound control limited – ISPs may ignore MED
TE with Aggregate: Prefer primary ISP with summarized routes; advertise specific prefixes on backup for fallover
Aggregate Impact: Longest-match wins → specific prefixes may steer traffic to alternate ISP; avoid provider-owned aggregates
RPKI – RESOURCE PUBLIC KEY INFRASTRUCTURE
– Framework to validate that a prefix is legitimately originated by a specific ASN
– Prevents route hijacking and accidental mis-originations (route leaks)
– RPKI validates origin → not the 5 RRs
KEY COMPONENTS
Trust Anchors (TA): Root CAs of the AS-Paths; issue certs for resource holders
ROA – Route Origin Authorization: Digitally signed object that authorizes ASN to announce a prefix (contains AS, prefix that AS can originate, max prefix length)
RPKI Validators: Software that downloads, verifies, and stores Validated ROA Payloads (VRPs)
RPKI VALIDATION STATES
Valid: Prefix + ASN match ROA
Invalid: Prefix found, but ASN mismatch or prefix too long
Not Found / Unknown: No matching ROA
VALIDATOR DETAILS
– Use TALS (Trust Anchor Locators) to fetch data from RIRs
– Validate cryptographic signatures (via X.509 certs with RFC 3779)
– Outputs VRPs; invalid objects are discarded
– Update frequency: ~24h (recommended), ~30–60min (practice)
– RRDP (RFC 8182) replacing rsync (uses HTTPS)
BGP MONITORING
Event tracking, BGP hijack detection, Route leak detection, RPKI status check, Reachability tracking, AS path change tracking, AS path visualization
NETWORK DESIGN
Pillars: Scalability, Speed, Availability, Security, Manageability → overall Cost
AVAILABILITY CONCEPTS (MOST IMPORTANT REQUIREMENT)
MTBF: Mean Time Between Failures; MTTR: Mean Time to Repair
$- A = \frac{MTBF}{MTBF + MTTR}$ $- \text{Lower MTTR} + \text{higher MTBF} = \text{more availability}$
REDUNDANCY
– Adds reliability, decreases MTBF but increases MTTR and complexity
– Balance: resilience vs. manageability
Backup Paths: Duplicate devices/links on primary path, build extra links for redundancy, consider backup link capacity, consider failover speed
Load Balancing: ECMP, EtherChannel, Port-Channel

HIERARCHICAL DESIGN
Access: Connect end devices, high port count, port security, L2, QoS marking
Distribution: L3, policy control, HSRP/VRP, loop protection, small fault domain
Core: High-speed backbone, L3 only, no policy, scalable/redundant, no security
Collapsed Core: Combines Core + Distribution (small/medium networks)
FLAT DESIGN
– Any-to-any; small networks, MPLS/LAN setups
– Lacks scalability and control
FABRIC DESIGN
Modern design using Underlay/Overlay
Underlay: transport (e.g., IP, MPLS)
Overlay: logical virtual topology (e.g., EVPN)
ENTERPRISE CAMPUS
– 100s/1000s of users, multiple buildings, one physical location
– Multiple interconnected LANs, connected via Ethernet and Wireless
FHRP – FIRST-HOP REDUNDANCY PROTOCOLS
– Ensures default gateway is always reachable, PCs can only have one
– Enables fast failover during router failure

VRP – Virtual Router Redundancy Protocol (Multivendor): Shared virtual IP, real MACs per router, Master router handles forwarding
HSRP – Hot Standby Router Protocol (Cisco): Shared virtual IP + virtual MAC, One active, others in standby → faster
GLBP – Gateway Load Balancing Protocol (Cisco): Load balancing + redundancy, Shared virtual IP + multiple virtual MACs, Roles: AVG: Answers ARP and sends virtual MAC addresses of AVFs AVF: Forwards traffic
DATA CENTER DESIGNS
ToR – Top of Rack: switches per rack, less cabling, easy expansions/ exchanges per 'rack', scalable glass fiber, ideal for high service density (full racks). But more switches, more ports, more L2 Srv-2 Srv traffic, more STP to be managed
EoR – End of Row: 1 switch per row, less switches, higher utilization of ports, switches all at one place, better L2 availability between racks. But more cabling
TRAFFIC DIRECTIONS
North-South: Between external networks (client-server, in/out DC)
East-West: Within DC (e.g., server-server, storage)
THREE-TIER DC
– Access – Aggregation – Core (like Hierarchical)
– Optimized for North–South traffic
– Not ideal for East–West communication
LEAF-SPINE ARCHITECTURE
– Two-tier: Leaf switches (access) connect to Spines (core)
– High performance, low latency
– Scalable, ideal for East–West traffic

MULTICAST
Use-cases 1-to-Many: Streaming, software updates, music-on-hold
Use-cases Many-to-Many: Gaming, VR, stock data, group chat
Benefits: Efficient bandwidth, lower server/CPU load, no redundancy, supports multipoit apps
Properties: UDP-based (no delivery guarantee, congestion control, or ordering) Apps must handle drops, duplicates, out-of-order packets
Source: Sends to group IP; doesn't need to join
Receiver: Must explicitly join group to receive traffic
MULTICAST ADDRESS RANGES
224.0.0.0 – 224.0.0.255: Link-local, TTL = 1 (not forwarded by routers)
224.0.1.0 – 224.0.1.255: Reserved by IANA, routable
224.0.0.0 – 239.255.255.255: Source-Specific Multicast (SSM)
239.0.0.0 – 239.255.255.255: Administratively scoped (private multicast space)
BROADCAST BASICS
L3 routes between subnets; L2 floods within same subnet
L2 (Bridging): MAC flit.filt.filt; switches flood to all ports in VLAN
L3 (Routing):
– 255.255.255.255: local broadcast, not routed
– Directed broadcast (e.g. 10.1.1.1.255): can be routed if enabled
L2 MULTICAST: MAC MAPPING
1) Get IP: Example multicast IP address: 239.5.5.5
2) Convert to Binary: 239.5.5.5 = 11011111.00000101.00000101.00000101
– Take only last 23 bits: 00000101.00000101.00000101
3) Map to MAC Prefix: Use fixed MAC multicast prefix: 01:00:5E → Map last 23 bits to: 01:00:5E:05:05:05
4) Final MAC Address: 01:00:5E:05:05:05
IGMP – INTERNET GROUP MANAGEMENT PROTOCOL (HOST TO FIRST-HOP-ROUTER)
Purpose: Manages group membership for IPv4 multicast on each segment

IGMPv1:
– Basic join via query-response mechanism
– No way for a host to leave a group explicitly
– Router sends general membership queries every 60s to 224.0.0.1
– If no report is received, router removes group after timeout
– Receiver has no knowledge of the multicast source
IGMPv2:
– Adds Leave Group message (faster pruning of unused traffic)
– General queries to 224.0.0.1 every 125s 'Any hosts interested in any groups?'
– Supports Group-Specific Queries e.g. when someone leaves group ('anyone still interested in group x?'), reducing broadcast overhead
– Still source-agnostic: receivers don't know who the source is
IGMPv3:
– Adds source filtering (Include/Exclude lists)
– Enables Source-Specific Multicast (SSM) – receiver requests traffic only from selected source(s), no need for Rendezvous Point (RP) anymore
– Adds support for applying source-level access control (source filtering)
– Can also be used in ASM (Any Source Multicast), but mainly with SSM

IGMP SNOOPING
Without snooping: multicast = broadcast on VLAN
With snooping: switch listens to IGMP messages and builds a forwarding table
Default: snooping is enabled; switch needs IGMP Query to operate
REVERSE PATH FORWARDING (RPF)
RPF Check: To avoid loops, verifies that a multicast packet arrives on the interface that a unicast packet destined for the multicast source would be forwarded out of.
RENDEZVOUS POINT (RP)
– Used in Shared Tree (*,G) setups with PIM-SM
– RP acts as the common meeting point for sources and receivers
– RPF check is performed toward the RP (not the source)
– Once the source is known, routers may switch to a Source Tree (S,G)
SHARED TREE VS. SOURCE-BASED TREE
Shared Tree (*,G):
– IGMP host sends a membership report (IGMP Join)
– Router adds (*,G) entry to multicast routing table
– * means 'any source' → source is unknown/unspecified
Source-Based Tree (S,G):
– Built when router receives an (S,G) join/report from IGMP host
– S = known multicast source; G = group
– Router adds (S,G) to mroute table once source is known

PIM – PROTOCOL INDEPENDENT MULTICAST (ONLY BETWEEN ROUTERS)
– Relies entirely on the unicast routing table (RIB) for multicast forwarding decisions
– Protocol-independent: works with static routes, OSPF, IS-IS, etc.
PIM-DH – DENSE MODE (NO RP)
Push Model: Floods multicast traffic to all interfaces; then prunes where no receivers exist
1) Flooding: Source sends traffic → forwarded out all multicast-enabled links using unicast RIB
2) Distribution Tree: Initially includes entire network (shared tree rooted at source)
3) Prune Messages: Routers without interested receivers send prunes upstream to remove themselves from the tree
4) State Maintenance: Routers track source, receivers, interfaces to forward/prune per group
PIM-SM – SPARSE MODE (ASM, SSM)
Push Model: Multicast traffic is only sent where requested. Works with IGMP to detect interested receivers and uses unicast routing for forwarding.
1) Join/Purge: Routers send explicit join/prune messages to request or stop receiving multicast for group (G) to other routers
2) Forwarding: Routers only forward multicast packets for group (G) on interfaces from which explicit joins were received
ASM – ANY SOURCE MULTICAST (HOW IT WORKS WITH RP)
– Works with IGMPv1 or IGMPv2 → receiver does not know the source
– Receiver sends IGMP Join (*,G) to its first-hop router
– First-hop router forwards PIM Join (*,G) hop-by-hop toward the Rendezvous Point (RP)
– RP acts as a common meeting point for sources and receivers
– Sources send multicast traffic to the RP via a PIM Register tunnel
– All routers in the multicast domain must know the RP location

Host Moved: When a host moves to a new VTEP, the new VTEP advertises updated reachability with a higher move sequence number. The old VTEP withdraws its entry, completing the migration.

ROUTE DISTINGUISHER (RD) VS. ROUTE TARGET (RT)

- **Route Distinguisher (RD):** Uniquely identifies VPN routes – allows same IP prefix to be used in different VPNs. Can be IPv4 or ASN *Used to make routes unique in BGP (VPNv4/v6)*. Forms VPNv4 NLRI: *RD:IPv4 prefix*
- **Route Target (RT):** Controls route import/export between VRFs: *Used as extended BGP community.*

How RTs Work:

- A route is tagged with an RT when advertised by BGP.
- Other VRFs import the route if the RT matches their import policy.
- Allows overlapping or shared connectivity between tenants (e.g., shared services).

– **Format:** Typically in the form *ASN:nn* or *IP:nn*, e.g., 65000:100, 1:100

- Multiple RTs can be used: A router can have multiple RTs for flexible policies (e.g., one RT for VPN, another for shared services)
- EVPN (ETHERNET VPN) - L2**
- KEY FEATURES**
 - L2 bridging across L3 networks
- BGP Control Plane:** Distributes MAC info (no flooding)
- VXLAN Overlay:** Encapsulates L2 in L3 UDP (data plane)
- Multi-Tenancy:** via VNI segmentation

Redundancy: All-active multihoming, ECMP, fast convergence

USE CASES

- Multi-tenant datacenter interconnects (DCI)
- Extending L2 over WAN between remote sites
- Scalable, segmented L2 fabrics

- PE's learn MACs from local CE's (data plane)
- MACs advertised via BGP (control plane)
- Uses Route Distinguishers and MPLS labels
- Remote PE's update L2 RIB/FIB with MAC and next-hop info
- Enables seamless L2 across IP/MPLS backbone

- EVPN uses MP-BGP with specific AFI/SAFI
- Supports multiple route types and attributes
- Unsupported routes are dropped by BGP

- Route Reflector (RR) avoids full-mesh iBGP
- RR reflects EVPN routes to other PEs
- RR doesn't participate in EVPN or pseudowires
- RR needs only **address-family l2vpn evpn**
- L2VPN RIB stores endpoint/VFI info for control plane
- **BGP UPDATE** from spines contain **ORIGINATOR ID** (origin leaf)

HOST DETECTION

- Host connects to VTEP → MAC learned locally
- VTEP advertises MAC + L2VNI via BGP EVPN
- MAC learning follows normal Ethernet semantics

INGRESS REPLICATION (IR)

- BUM traffic, when Multicast underlay network is not used, handle multi-destination traffic (ARP → unicast)

EARLY ARP TERMINATION (ARP SUPPRESSION)

Avoids flooding ARP requests

- VTEP queries control plane for MAC/IP/VNI mapping
- If known → direct unicast (no broadcast)

SILENT HOST FLOW (FALLBACK)

- If IP/MAC unknown → ARP sent via **ingress replication**
- Replicated ARP request goes to remote VTEPs

- Only correct host responds → update reflected to all VTEPs
- Future traffic uses updated BGP mapping

VRF - VIRTUAL ROUTING AND FORWARDING

- Multiple isolated routing tables on one device
- Each tenant = one VRF → traffic isolation

- Supports independent policies per tenant
- Key for scaling and multi-customer separation

IRB - INTEGRATED ROUTING AND BRIDGING

- Enables inter-VLAN routing inside EVPN

- Avoids central gateway → no 'traffic tromboning'
- Two modes: Symmetric and Asymmetric

SYMMETRIC IRB (L2 + L3)

- Routing/bridging on ingress + egress VTEPs
- Uses L3 Transit VNI (same in both directions), One L3 VNI per VRF (Tenant)

- Scales well, clean separation of MAC and IP
- **ASYMMETRIC IRB (L2)**
 - Routing only on ingress, bridging on egress
 - VXLAN uses destination VNI in both directions
 - One L2 VNI per VLAN/Subnet
 - Simple config, but requires all VLANs/VNIs on all VTEPs

- Same gateway IP+MAC on all VTEPs
- Enables local default gateway for hosts
- Supports mobility + optimal forwarding

- Host sends ARP/ND to local VTEP
- VTEP learns MAC/L2VNI and IP/L3VNI
- Info is advertised in EVPN (control plane)

- Label Switched Path (LSP) → pre-determined path across MPLS network
- advantage eBGP between PE-CE: No mutual redistribution, same routing process
- encrypt traffic flowing over MPLS L3VPN backbone? yes (e.g. bank)
- Unicast Reverse Path Forwarding (uRPF): checks source of each packet & verifies that source is in routing table

- control plane (e.g. OSPF) → to learn labels
- iBGP used to exchange NLR (RD, RT, IPv4 Prefix, NextHop & VPN Label) between PE
- imp-null = networks are directly connected, no more label switching

WAN

Connects remote LANs via SPs for data/voice/video; key needs: bandwidth, control, design, resilience, mgmt. **Requirements:**
Bandwidth: App needs, peak usage, reserve for VoIP
Control/Security: Trust provider? No full control
Availability: Redundancy, SLA for failures

- SR Control Plane Segment Types**
 - IGP Prefix Segment:** Global SID tied to IGP prefix (multi-hop); all nodes install forwarding entries
 - IGP Node Segment:** Global SID for a specific node (shortest-path forwarding)
 - IGP Anycast Segment:** Global SID for a group of nodes; traffic sent to nearest
 - IGP Adjacency Segment:** Local SID; direct link to neighbor
 - L2 Adjacency SID:** Local SID for Layer-2 segment (e.g., Ethernet link)
- Combining Segments**
 - End-to-end paths can mix IGP and BGP segments
 - Traffic to BGP Anycast → more ECMP in data centers
- SR-NPLS**
 - Reuses existing MPLS data plane — no hardware change needed

- Supports both IPv4 and IPv6 networks

BENEFITS OF SEGMENT ROUTING

Benefits: Simplification (removes protocols, simple operations, admin and mgmt), enhanced Traffic eng. (Delay, Bandwidth, Packet Loss, TE metric: Controller, Source-Node), Seamless deployment, Robust, Network Innovation (SDN, Cloud Native Networking)

Source Router: Balances distributed intelligence with centralized optimization

TI-LFA: Fast reroute techniques; protects against link/node failure with micro-loop avoidance and no pre-calculation dependency

Traffic Engineering (TE): Optimizes network performance by analyzing and controlling traffic to reduce congestion and improve QoS

Service Function Chaining (SFC): Chains SDN services in order; automates traffic between VNFS and optimizes routing for performance

QoS
Internet is best effort: no guarantees, no QoS; all traffic treated equally (net neutrality); simple, scalable, but no delivery/order assurance or prioritization
QOE & ROUTE PINNING
QOE – Quality of Experience: Perceived service quality from user perspective

Route Pinning: Keeps flow on a fixed path to prevent oscillation (don't switch immediately to "better" path)

NETWORK PERFORMANCE METRICS

Latency / Delay [ms]: Time for packets to travel src → dest (Voip < 150ms)

End-to-End Delay: Total time needed to receive

Jitter [ms]: Variation in delay between packets, caused by re-routing/queuing (Voip<30ms). Calc: no queue - queued delay

Throughput: Rate of successfully delivered data
 Packet Loss [%]: Dropped packets due to congestion or errors (Voip < 1%)
 Bandwidth [Gbit/s]: Maximum transfer capacity of a link

QUEUEING ALGORITHMS

FIFO (First-In First-Out): Basic, no prioritization

Priority Queuing (PQ): Multiple queues, serve highest first; others may starve
Round-Robin: One packet per queue in turn (fair, but ignores priority)
Weighted Fair Queuing (WFQ): Round-Robin with weights, e.g., 2 packets from Q1, 4 from Q2
Class-Based WFQ (CBWFQ): WFQ with user-defined classes, queue limits max bandwidth guaranteed or max % of bandwidth (logical queues based on traffic class)

Low Latency Queuing (LLQ): Adds strict priority queue (priority class) to CBWFQ for delay-sensitive traffic (e.g. voice) (based on IP Precedence, DSCP, src, port, protocol...)

Tail Drop: Drops packets when queue full; huge interruption of traffic → same as no connectivity
TCP Global Sync: Many TCP flows back off and restart simultaneously → link underutilization
TCP Starvation: TCP slows down after drops. UDP doesn't → queues filled

RED: Random early drops before full queue to prevent global sync and TCP collapse. Dropped TCP segments cause TCP sessions to reduce their window sizes

WRED: RED + DSCP/EXP-based drop logic, prioritizes higher-marked traffic

DSCP / EXP: DSCP (6-bit in IP header) marks packets for QoS; used in Diff-Serv for classifying traffic. EXP (3-bit in MPLS label) serves same purpose elsewhere within MPLS networks; often mapped from DSCP.

POLICING VS. SHAPING

Policing (Inbound mostly): Drops packets that exceed configured rate limits

Shaping (Outbound): Buffers packets to smooth traffic bursts and conform to profile

QOS MODELS

Best Effort: No guarantees, all traffic treated equally (follows Internet neutrality)

Integrated Services (IntServ): End-to-end QoS, per-flow resource reservation, precise but not scalable (uses RSVP)

Differentiated Services (DiffServ): Class-based, scalable approach using marking (e.g.,), no hard guarantees

TRAFFIC MARKING

L3 Marking: ToS byte → DSCP (6 bits) + IP Precedence (3 bits)
L2 Marking: Dot1q header → 802.1p CoS bits

MODULAR QOS CLI (MQC)

Class Map: Define traffic classes (e.g., match voice or video)
Policy Map: Define actions for each class (e.g., limit, shape, priority)

Service Policy: Apply policies to interfaces or directions (in/out)

CDN

Origin Server: Central content source (original files), usually in a datacenter

Edge / CDN Server (POP - Point of Presence): Geographically distributed caches content

DNS Infrastructure: Directs users to optimal edge server (e.g. via Geo-Routing)

KEY BENEFITS

- Latency Reduction:** Nearby edge servers reduce round-trip time
- Availability:** Failover and redundancy in case of node failure

- Scalability:** Handles traffic spikes via load balancing
- Cost Optimization:** Reduces backend and transit load on origin
- DDoS Protection:** Edge servers absorb attacks → not all traffic on one server
- Global Load Reduction:** Less long-distance traffic across the Internet

Pros: Fast failover, simple, no app logic needed
Cons: Less control, BGP != best latency, route flapping risk

HTTP CACHING & HEADERS

- Caching is controlled via HTTP headers between clients, proxies, and servers

ETag: Validator tag (version/hash), used with *If-None-Match*
Last-Modified: Timestamp used with *If-Modified-Since* for revalidation
Age: Time (in seconds) since response was fetched from origin
Validation: Client uses *ETag* or *Last-Modified*; server returns 304 if unchanged

EVPN BGP ROUTING TABLE INFO
 = Would not be there if it was L2 VNI BGP Routing Table
Route Distinguisher: 172.16.255.101:32777
Route Type: 2
MAC Address Length: 48

```
# nbf-03 show bgp l2vpn evpn 10.10.8.100
BGP routing table information for VRF default, address family L2VPN
VRF
Route Distinguisher: 172.16.255.101:32777
BGP routing table entry for
[Prefix] (1 available) best #1
[0]: [10.10.8.100/24] (N524,0089,29a8):[32]:[10.10.8.100]/272, version 19897
[OrigID]: (9x00E022) (5154,00000000) on xmit-list, is not in L2rrib/evpn,
is not in HW
advertised path-id 1
Path type: internal, path is valid, is best path, no labeled nexthop
Imported to 2 destination(s)
AS-Path: NONE, path sourced internal to AS
172.16.254.101 (metric 81) from 172.16.255.1 (172.16.255.1)
received IGP, MED not set, localpref 180, weight 6
received Label 30010 50800
Outgoing community: RT:1;RT:65000;50000 EBNP AC: Router MAC:5254.08ca.69ae
Originator: 172.16.255.101 Cluster List: 172.16.255.1
```

PRÜFUNG VORJAHR

NETWORK DESIGN

3-tier campus network: Default Gateway (D), QoS marking (A), STP Root Port (A), HSRP VRRP or GLBP (D), "Simple" (C), OSPF Totally Stub Area

- VXLAN is a data plane technology which encapsulates Ethernet frames in UDP datagrams to tunnel layer 2 frames over a layer 3 network.
- The underlay network is unaware of VXLAN devices that connect to the physical switches are unaware of VXLAN.
- A route distinguisher is used to uniquely identify a route in combination with